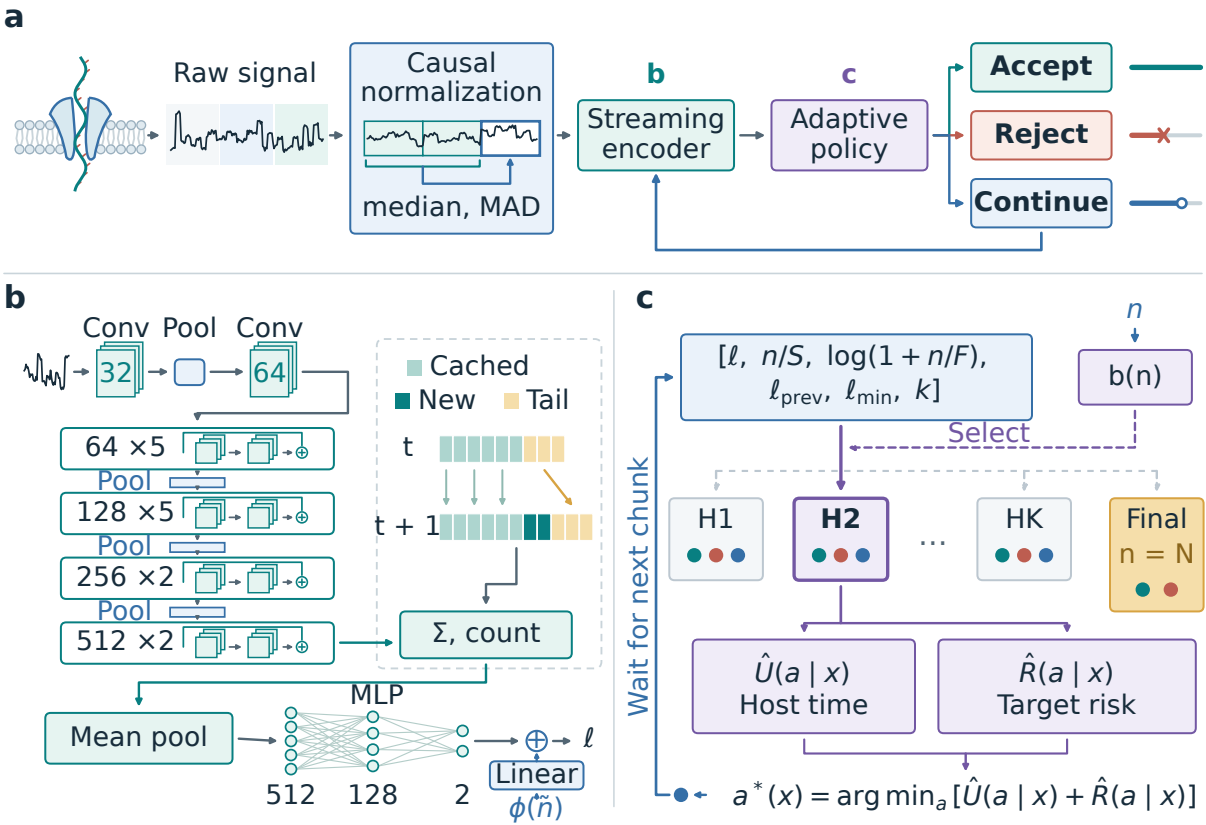
a
Raw signal
Causal normalization
median, MAD
b
Streaming encoder
c
Adaptive policy
Accept
Reject
Continue
b
Conv
Pool
Conv
32
64
64 ×5
Pool
128 ×5
Pool
256 ×2
Pool
512 ×2
Cached
New
Tail
t
t + 1
Σ, count
MLP
Mean pool
512
128
2
Linear
$\phi(\tilde{n})$
ℓ
c
n
$[\ell,\ n/S,\ \log(1+n/F),$
$\ell_{\text{prev}},\ \ell_{\text{min}},\ k]$
b(n)
Select
H1
H2
...
HK
Final
n = N
$\hat{U}(a \mid x)$
Host time
$\hat{R}(a \mid x)$
Target risk
Wait for next chunk
$a^*(x) = \arg\min_a [\hat{U}(a \mid x) + \hat{R}(a \mid x)]$